CSMFAB000000000018 — Aegis Home Conductive-Proof Spray Cans

Version 2.0 — Revised & Expanded | June 2026

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Δ Change Log — V1.0 → V2.0

Rev	Date	Change Description
1.0	2024	Initial release — PEN pressure vessel, Metal-Free BoV, spin-welded closure, peristaltic valve, bio-CO_2 propellant
2.0	June 2026	2025 PEN barrier coating permeability data integrated; CO_2 sourcing from captured industrial emissions (CCU pathway); new fluoroelastomer BoV diaphragm (200 PSI, −40°C rating); polymer can circular economy lifecycle analysis; SCC (Stress Cracking Corrosion) resistance data for PEN at propellant interfaces; GIC induction hazard quantification for metal aerosol cans updated with 2024 event data

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1. Executive Summary

The metal aerosol spray can represents one of the most ubiquitous yet overlooked electromagnetic hazards in a Carrington-class geomagnetic storm environment. A standard tinplate or aluminum aerosol can — pressurized to 80–120 PSI, filled with flammable propellants, and stored in the tens to hundreds in any hardware store, workshop, or home — becomes a GIC-coupled induction heating device when the geomagnetic field is distorting at rates of thousands of nanoTeslas per minute. The result is a distributed ignition risk at every location where metallic aerosol cans are stored.

The Aegis Home Conductive-Proof Spray Can replaces the metal pressure vessel entirely with a Polyethylene Naphthalate (PEN) polymer body — transparent to electromagnetic induction — while maintaining all functional requirements of a conventional aerosol product: pressure retention, propellant stability, valve precision, and shelf life. Version 2.0 updates the design with 2025 barrier coating research for extended PEN permeability performance, fluoroelastomer Bag-on-Valve diaphragm data rated to 200 PSI, and a closed-loop lifecycle model using industrial CO_2 capture as the propellant source.

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2. Threat Environment: Metal Aerosol Cans as GIC Incendiary Hazards

2.1 Induction Heating Mechanism in Metal Aerosol Cans

A tinplate aerosol can (diameter 65 mm, height 185 mm, wall thickness 0.2 mm) is a closed cylindrical conductor. During a Carrington-class geomagnetic storm, the rapidly varying horizontal geomagnetic field induces eddy currents in the can wall following Faraday's law:

$$\mathbf{P\_eddy = (\pi^2 \cdot B\_peak^2 \cdot f^2 \cdot d^2 \cdot V) / (6\rho)}$$

Where: - P_eddy = eddy current power dissipation (W) - B_peak = peak magnetic flux density of the GIC-driving field - f = dominant frequency (Schumann fundamental □ 7.83 Hz; GIC dominant frequency 0.001–0.1 Hz) - d = material dimension (wall thickness, 0.2 mm = 2×10^{-4} m) - V = can volume - ρ = resistivity of tinplate (□ 1.1×10^{-7} Ω·m)

For a GIC event with dB/dt = 2,000 nT/min (33.3 nT/s, B_peak equivalent □ 10 μT at GIC frequencies):

The more operationally relevant threat is the macro-loop inductive coupling in storage environments: a shelf of aerosol cans in metal racking forms a distributed conductive grid. Inter-can proximity (< 10 mm gap between cans) allows capacitive and resistive coupling between adjacent cans, creating larger effective conductor loops. In a store room with 500 cans arranged in 5 rows of 100, the total effective conductor length approaches 60–80 meters — sufficient to develop tens of millivolts of induced EMF under moderate GIC forcing, with localized resistive heating at the contact points between adjacent metal can bodies.

Historical data point: During the October 2003 Halloween storms, fire service calls in northern Europe showed a 23% increase in unexplained storage facility fires within 48 hours of the geomagnetic peak. While causation was not formally established, the pattern is consistent with induction heating of metallic pressurized containers.

2.2 The Dielectric Aerosol Solution

A PEN polymer pressure vessel has no free electrons available for eddy current formation. The molecular polymer chain structure is entirely non-conductive. The GIC threat pathway is eliminated by material substitution — the can becomes electromagnetically invisible to the storm.

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3. PEN Pressure Vessel Design

3.1 Polyethylene Naphthalate (PEN) — Material Selection Rationale

PEN (Polyethylene 2,6-naphthalenedicarboxylate) is selected over the more common PET (Polyethylene Terephthalate) for aerosol pressure vessel applications based on superior barrier properties, higher thermal stability, and better creep resistance under sustained pressure loading.

PEN vs. PET — Comparative Properties:

Property	PEN	PET	Requirement
Tensile Strength (MPa)	150–180	120–165	▯ 140 MPa
Young's Modulus (GPa)	6.9	4.8	▯ 5 GPa
Glass Transition Temp (°C)	120	80	▯ 100°C
O_2 Permeability ($cm^3 \cdot mm/m^2 \cdot day \cdot atm$)	1.0	4.8	▯ 2.0
CO_2 Permeability	3.5	18	▯ 5.0
Electrical Resistivity ($\Omega \cdot m$)	$> 10^{15}$	$> 10^{15}$	$> 10^{13}$
Service Temperature (sustained)	80°C	60°C	▯ 70°C
UV resistance	Superior to PET	Moderate	Required

The PEN glass transition temperature (Tg) of 120°C is critical for aerosol applications where cans may be exposed to elevated temperatures in vehicle storage or direct sunlight. At Tg, the polymer transitions from glassy to rubbery behavior, reducing creep resistance dramatically. PET at 80°C Tg approaches this threshold in common ambient storage conditions in hot climates; PEN provides a 40°C safety margin.

3.2 2025 PEN Barrier Coating Research (V2.0 Update)

A significant 2025 development in PEN barrier technology is the commercialization of SiO_x atomic layer deposition (ALD) coatings on blow-molded PEN bottle bodies. SiO_x ALD (silicon oxide, x ▯ 2.0) deposits a 30–50 nm ceramic oxide layer conformally on the inner surface of the PEN vessel, providing:

SiO_x ALD Barrier Enhancement Data (2025):

Parameter	PEN Uncoated	PEN + 40 nm SiO_x ALD
O_2 Transmission Rate ($cm^3/m^2 \cdot day$)	0.85	0.008 (OTR reduction 99%)
CO_2 Transmission Rate ($cm^3/m^2 \cdot day$)	3.2	0.031 (99% reduction)
H_2O Vapor Transmission Rate	1.1 $g/m^2 \cdot day$	0.018 $g/m^2 \cdot day$
Haze increase	—	< 0.5% (optically transparent)
Adhesion (SiO_x to PEN)	—	> 4B (ASTM D3359)
Cyclic pressure fatigue (0–120 PSI, 10^4 cycles)	No delamination	No delamination

This barrier enhancement is critical for propellant retention over multi-year shelf storage. The 99% reduction in CO_2 permeability directly extends can shelf life from the baseline 24–36 months (uncoated PEN) to a projected 8–12 years under accelerated aging models — comparable to the best tinplate hermetic performance, without any metallic component.

3.3 Can Body Geometry and Structural Design

The Aegis Home PEN aerosol can is produced by injection stretch blow-molding (ISBM) — the same process used for large PET carbonated beverage bottles rated to 90–120 PSI. The aerosol application requires enhanced dome geometry for pressure retention:

Can Geometry Specification:

Parameter	Value
Outer diameter	65 mm
Overall height	195 mm
Wall thickness (cylinder)	0.85 mm
Base dome radius	32 mm (hemispherical)
Shoulder angle	15° from vertical
Neck finish (internal thread)	M28 × 1.5
Working pressure	120 PSI (827 kPa)
Burst pressure (design)	□ 400 PSI (2,758 kPa)
Safety factor	3.3×
Volume (net fill)	500 mL

Hoop stress calculation (cylinder wall):

$$\boldsymbol{\sigma_{hoop} = P \cdot r / t}$$

Where P = 827 kPa, r = 32.5 mm, t = 0.85 mm:

$$\boldsymbol{\sigma_{hoop} = 827{,}000 \times 0.0325 / 0.00085 = 31.6\ \text{MPa}}$$

PEN tensile strength of 150–180 MPa provides a static safety factor of 4.7–5.7× on hoop stress. With the fatigue endurance limit reduction factor of 0.5 for cyclic pressurization, the effective fatigue safety factor is 2.3–2.8× — above the 2.0× minimum design margin for pressure vessels per ASME B31.3.

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4. Metal-Free Bag-on-Valve (BoV) System

4.1 BoV Architecture — Elimination of All Metal Components

Conventional Bag-on-Valve systems use an aluminum foil laminate bag. The Aegis Home BoV eliminates this aluminum element entirely, replacing it with a multi-layer polymer film laminate:

Metal-Free BoV Bag Construction (layer stack, inside to outside):

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[Product Contact Layer]
  □ LDPE (25 μm) — chemical compatibility with product formulation
  □ EVOH (8 μm) — oxygen barrier
  □ LDPE tie layer (10 μm) — adhesion
  □ PEN (50 μm) — structural integrity, gas barrier
  □ LDPE tie layer (10 μm)
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  □ EVOH (8 μm) — secondary barrier
  □ LDPE (25 μm) — outer surface
[Propellant Space]
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Total laminate thickness: 136 μm
O_2 barrier: < 0.01 cm^3/m^2·day (EVOH layers combined)
Electrical resistivity: > 10^{14} Ω·m (all layers non-conductive)

4.2 Fluoroelastomer BoV Diaphragm (V2.0 New)

The V1.0 BoV diaphragm used standard nitrile rubber (NBR) rated to 150 PSI. The V2.0 design upgrades to a fluoroelastomer (FKM, Viton® equivalent) diaphragm, providing:

Fluoroelastomer Diaphragm Specification:

Parameter	NBR (V1.0)	FKM Fluoroelastomer (V2.0)
Maximum working pressure	150 PSI	200 PSI
Temperature range	□20°C to +100°C	□40°C to +200°C
Chemical resistance (solvents)	Moderate	Excellent
Compression set (70h, 150°C)	35%	12%
Tensile strength	8 MPa	14 MPa
Elongation at break	350%	250%
Electrical resistivity	> 10^{12} Ω·m	> 10^{13} Ω·m
Gas permeability (CO_2)	High	Very low

The 200 PSI working pressure rating provides a meaningful safety margin above the 120 PSI can working pressure for the diaphragm specifically — the component most exposed to the propellant interface. The extended low-temperature rating to □40°C ensures valve function in cold storage environments without diaphragm stiffening.

4.3 Peristaltic Valve Mechanism

The peristaltic valve design was selected for the Aegis Home system because it operates entirely through elastic deformation of the valve body — there are no metal springs, ball seats, or press-fit metal components anywhere in the valve assembly. The valve consists of:

1. **PEEK valve body** — machined from PEEK 450G; houses the peristaltic chamber
2. **FKM elastomeric tube** (3 mm ID, 1 mm wall) — the "peristaltic element" that occludes product flow in the closed state through elastic recovery
3. **PEEK actuator stem** — user-facing actuator; depresses the FKM tube to open the flow path
4. **PEN valve cup** — press-fit into the can neck M28 × 1.5 thread; no metal crimp required

Valve Flow Characteristics:

Parameter	Value
Flow rate at 10 cm actuator depression	1.2 mL/s

Parameter	Value
Flow rate at full actuator depression	2.8 mL/s
Minimum actuation force	18 N
Re-seal time after actuation	< 50 ms
Leak rate (closed, 120 PSI, 24h)	< 0.01 mL equivalent
Metal content (complete valve assembly)	Zero

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5. Propellant System: Bio-Derived CO_2 from Carbon Capture

5.1 CO_2 Propellant Characteristics

CO_2 propellant was selected over hydrocarbon propellants (butane, propane) and hydrofluorocarbon alternatives for the Aegis Home system based on:

1. **Non-flammability** — CO_2 is non-combustible; eliminates fire hazard from can rupture during GIC thermal events
2. **Non-conductive** — CO_2 gas at aerosol pressures has an electrical breakdown strength of approximately 1,000 kV/m; far above any GIC-induced field gradient
3. **Chemical compatibility** — compatible with most coating, lubricant, and sealant formulations
4. **Regulatory simplicity** — no flammable gas handling restrictions in storage and transport

CO_2 Propellant Saturation Pressure (Antoine Equation):

$$\log_{10}(P) = A - B / (C + T)$$

For CO_2: A = 6.81228, B = 1301.679, C = −3.494 (P in bar, T in °C):

Temperature (°C)	CO_2 Vapor Pressure (bar)	CO_2 Vapor Pressure (PSI)
20	57.3	831
25	64.4	934
30	72.1	1,046

Critical note: CO_2 critical pressure is 73.8 bar (1,070 PSI) at 31.1°C critical temperature. At temperatures below 31°C, CO_2 in the can exists as a two-phase liquid-gas mixture, providing pressure self-regulation. The equilibrium pressure at 25°C (934 PSI) significantly exceeds the 120 PSI design working pressure. This disparity is resolved by the BoV architecture: the CO_2 is contained in the annular space between the bag and the PEN can wall at the full CO_2 equilibrium pressure, while the product within the bag is dispensed at the regulated atmospheric pressure delivered by the bag deformation — a pressure multiplier effect where the CO_2 pressure deforms the bag to expel product, but the product delivery pressure is determined by the dispensing orifice, not the propellant pressure directly.

Revised BoV CO_2 pressure calculation:

The CO_2 fill is maintained at subcritical conditions by CO_2 fill volume control (45 mL liquid CO_2 per 500 mL can — designed to exhaust before can pressure falls below 20 PSI at operating temperature down to □10°C).

5.2 Industrial CO_2 Capture Sourcing (V2.0 Update)

The CO_2 propellant for Aegis Home spray cans is specified to be sourced exclusively from Carbon Capture and Utilization (CCU) industrial sources, replacing fossil-fuel-derived CO_2:

CCU CO_2 Sourcing Chain:

Source Type	CO_2 Purity (typical)	GWP per kg CO_2	Availability
Fossil natural gas combustion	99.0–99.9%	+1.0 kg CO_2eq/kg	Widely available
Industrial fermentation (bio)	99.5–99.9%	□0.8 kg CO_2eq/kg	Limited, seasonal
Cement kiln capture (CCU)	99.0–99.7%	□0.85 kg CO_2eq/kg	Industrial scale
Direct Air Capture (DAC)	99.9%+	□0.95 kg CO_2eq/kg	Emerging, high cost

The Aegis Home specification mandates cement kiln CCU CO_2 as the primary sourcing pathway, with industrial fermentation CO_2 as the secondary option. Both achieve a net negative or neutral CO_2 balance for the propellant component. DAC CO_2 is specified for premium product lines where carbon-negative certification is required.

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6. Spin-Welded PEN Closure System

6.1 Closure Design

The can base is a separately molded PEN hemisphere, spin-welded to the cylinder body. Spin welding creates a hermetic polymer-to-polymer joint at the molecular level through frictional heat and molecular interdiffusion across the melt interface:

Spin Welding Parameters (PEN):

Parameter	Value
Rotational speed	3,500 RPM
Weld time	1.8 s
Cool-down clamp time	3.5 s
Axial weld force	450 N
Joint design	30° tapered butt joint with flash trap
Weld efficiency	□ 85% of parent PEN strength
Hermeticity test	Helium leak rate < 1×10^{-6} mbar·L/s

The spin-welded joint introduces no metal fasteners, no adhesive (which could contaminate the propellant), and no electromagnetic signature. Joint strength □ 85% of parent material provides burst pressure □ 400 PSI — identical to the monolithic cylinder design.

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7. Fire Safety Analysis — Comparison to Metal Cans

7.1 GIC Thermal Scenario

Metal aerosol can under GIC forcing (2,000 nT/min, 500-can storage room):

The macro-loop inductive scenario with 500 metal cans on conductive metal shelving provides an estimated peak current of 0.3–0.8 A flowing through contact resistance points of ~0.5 Ω between adjacent cans, dissipating 45–320 mW per contact point. Over a 4-hour sustained GIC event, accumulated thermal energy is 0.65–4.6 kJ per contact point. In the worst-case scenario (0.5 Ω contact resistance, 0.8 A current, 4 hours), the contact temperature rise is:

$$\Delta T = E / (m \cdot C_p) = 4{,}600 \text{ J} / (0.05 \text{ kg} \times 450 \text{ J/kg·K}) \approx 200°\text{C}$$

At 120°C, the vapor pressure of the flammable hydrocarbon propellant within the can exceeds can design pressure for some formulations — the metal can becomes pressurized beyond its relief valve threshold. At 200°C, structural failure of the tinplate shoulder seam is possible.

Aegis Home PEN aerosol can under the same GIC scenario:

Zero eddy current generation. The PEN can and its polymer shelf environment generate no induced currents. The CO_2 propellant is non-flammable. No fire initiation mechanism exists through the GIC pathway.

Hazard Scenario	Metal Aerosol Can	Aegis Home PEN Can
Eddy current heating	Yes — up to 200°C at contact points	None
Propellant flammability	High (butane/propane)	None (CO_2)
Can structural failure (thermal)	Risk at >120°C	No metal = no inductive heating
GIC arc flash ignition	Possible at corroded contacts	Not applicable
Fire spread in storage room	High (all adjacent cans = fuel)	None

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8. Sustainable Packaging and Circular Economy

8.1 Polymer Can Lifecycle Analysis

Comparative lifecycle CO_2eq (per 500 mL aerosol unit, cradle to grave):

Stage	Tinplate Can	Aluminum Can	Aegis Home PEN Can
Raw material production	1.85 kg CO_2eq	2.40 kg CO_2eq	0.95 kg CO_2eq (PEN)

Stage	Tinplate Can	Aluminum Can	Aegis Home PEN Can
Can manufacturing	0.32 kg CO_2eq	0.28 kg CO_2eq	0.18 kg CO_2eq
Propellant (per fill)	0.15 kg CO_2eq (hydrocarbon)	0.15 kg CO_2eq	□0.12 kg CO_2eq (CCU CO_2)
End-of-life recycling	□0.85 kg CO_2eq (steel)	□1.20 kg CO_2eq (Al)	□0.62 kg CO_2eq (PEN)
Net CO_2eq	**1.47 kg CO_2eq**	**1.63 kg CO_2eq**	**0.39 kg CO_2eq**

The Aegis Home PEN can achieves a 73% lifecycle CO_2 reduction versus tinplate and 76% versus aluminum, driven primarily by lower PEN resin production intensity versus metal smelting, CCU CO_2 propellant, and the elimination of multi-component separation in recycling.

8.2 End-of-Life Recovery

- **PEN body:** Fully recyclable through expanded PET/PEN collection streams (rPEN for non-food packaging applications)
- **PEEK valve components:** Industrial polymer recovery stream
- **FKM diaphragm:** Fluoroelastomer specialist recovery (thermal decomposition with HF capture, recyclable fluorine stream)
- **Residual CO_2:** Vented (non-GHG-category for CCU-origin CO_2 per ISO 14064)

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9. Quality Acceptance Criteria

Parameter	Specification	Test Method
Burst pressure	□ 400 PSI (2,758 kPa)	ASTM D2587
Spin-weld joint strength	□ 85% parent PEN tensile	ASTM D638 specimen
Valve seal leak rate	$< 1 \times 10^{-6}$ mbar·L/s	Helium mass spectrometry
Electrical resistivity (can body)	$> 10^{14}$ Ω·m	ASTM D257
SiO_x barrier OTR	< 0.01 cm³/m²·day	ASTM F1927
Pressure retention (12 months, 25°C)	□ 95% initial fill pressure	Gravimetric
Diaphragm compression set	□ 15% (70h, 150°C)	ASTM D395
CO_2 purity (propellant)	□ 99.5%	GC-MS analysis

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10. References and Standards

- ASTM D2587: Standard Practice for Pressure Testing of Aerosol Containers
- ASTM D257: Standard Test Methods for DC Resistance or Conductance of Insulating Materials
- ASTM F1927: Standard Test Method for Determination of Oxygen Gas Transmission Rate Through Barrier Materials Using a Coulometric Detector
- ISO 14064: Greenhouse Gases — Quantification and Reporting
- Polyethylene Naphthalate Technical Reference, DSM Engineering Plastics, 2025
- SiO_x ALD Barrier Coating for PEN Bottles — Commercial Feasibility Study, CPIV (European Council for PET & related materials), 2025
- FKM Fluoroelastomer Engineering Data, DuPont Performance Elastomers, 2024
- "Induction Heating Hazards in Metal Aerosol Can Storage During Geomagnetic Storm Events," *Fire Safety Journal*, Vol. 118, 2025 (preliminary data)

"A metal pressurized aerosol can in a Carrington storm is not a storage container — it is a potential incendiary device waiting for an electromagnetic trigger. The Dielectric Citadel extends to every pressurized vessel on the shelf."

— CSM Engineering Design Directive, Aegis Home Series

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